Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

Abstract

Large language models (LLMs) are increasingly used to support literature discovery, synthesis, summarization, and the preparation of research reviews. Their usefulness, however, is constrained by a fundamental temporal asymmetry: scientific knowledge evolves continuously, whereas a conventional pretrained model acquires most of its parametric knowledge from a finite historical training corpus. This temporal mismatch is particularly consequential in rapidly evolving fields such as artificial intelligence, biomedicine, climate science, cybersecurity, and emerging technologies, where important findings, terminology, datasets, methods, and consensus positions may change substantially within months. This paper examines how knowledge cutoffs affect LLM-generated literature reviews and argues that the problem extends beyond simple omission of recent papers. Cutoffs can produce systematic distortions in literature coverage, citation selection, interpretation of research trends, assessment of methodological consensus, and identification of research gaps. The paper synthesizes research on temporal knowledge in LLMs, retrieval-augmented generation (RAG), continual learning, knowledge editing, hallucination, citation reliability, and automated scientific literature synthesis. Current approaches—including web retrieval, scholarly RAG, temporal filtering, continual updating, model editing, agentic research systems, and citation verification—substantially reduce some cutoff effects but do not eliminate temporal misalignment. In particular, retrieval systems can themselves contain stale or conflicting information, while LLMs may fail to distinguish historical from current evidence. Recent benchmarks demonstrate persistent weaknesses in temporal awareness and outdated-information handling, while emerging systems such as OpenScholar demonstrate the potential of specialized retrieval and iterative verification. The paper identifies a central research gap: literature-review systems require evaluation not merely for factuality and citation correctness, but for temporal completeness, temporal attribution, trend fidelity, and reproducibility. Future systems should therefore treat time as a first-class dimension of scholarly evidence, combining timestamp-aware retrieval, source provenance, temporal reasoning, claim-level citation verification, uncertainty estimation, and human oversight.

Keywords: large language models; knowledge cutoff; literature review; scientific literature synthesis; temporal knowledge; retrieval-augmented generation; citation hallucination; continual learning; research agents; temporal alignment

1. Introduction

Literature reviews are fundamental to scientific research because they organize existing evidence, identify methodological developments, expose disagreements, and establish the rationale for new investigations. Traditionally, preparing a high-quality review requires iterative database searching, screening, reading, comparison, citation management, and expert synthesis. The growth of scientific publication has made this process increasingly difficult for individual researchers. Recent work on automated literature research explicitly identifies the volume of scientific publications as a major bottleneck for human researchers and motivates LLM-based review-generation systems as a means of reducing this cognitive burden (Wu et al., 2025).

LLMs have changed this workflow by enabling researchers to ask questions in natural language and receive coherent summaries, comparisons, explanations, and draft reviews. However, conventional LLMs are not continuously synchronized with the scholarly record. Their parametric knowledge is primarily acquired

during pretraining, and therefore a model with a training cutoff cannot inherently know papers, discoveries, terminology, or empirical results that emerged afterward. Retrieval-augmented generation (RAG) was introduced partly to address this limitation by combining parametric model knowledge with an external non-parametric memory that can be updated independently of model weights (Lewis et al., 2020).

The knowledge-cutoff problem is more complicated than simply asking whether an LLM "knows" a recent paper. Cheng et al. (2024) demonstrate that the effective cutoff of an LLM can differ from its officially reported cutoff because training corpora contain documents with different temporal properties. Older material can remain present in newer web crawls, while deduplication procedures can create additional temporal inconsistencies. Thus, a model's temporal knowledge should be understood as a distribution rather than as a single date.

For literature reviews, this distinction is particularly important. A review is not merely a collection of facts. It is a structured representation of how a field has developed. If an LLM misses recent studies, it can misrepresent the current state of evidence, incorrectly identify an old problem as unsolved, fail to recognize a newly established method, overstate the importance of obsolete techniques, or describe a historical research debate as if it were current. Consequently, knowledge cutoff effects can become epistemic distortions rather than ordinary factual errors.

This paper examines these effects systematically. It focuses on five questions:

1. How do knowledge cutoffs influence LLM-generated literature reviews?
2. Why are rapidly evolving fields especially vulnerable?
3. Which technical approaches currently mitigate temporal limitations?
4. What new failure modes arise when retrieval and agentic systems are introduced?
5. What evaluation and architectural principles are required for temporally reliable literature-review systems?

The paper is a conceptual and narrative synthesis of scholarly research available through August 2026. It does not claim to constitute a systematic review of every publication on LLM-assisted literature review. Instead, it integrates research from LLM temporal reasoning, RAG, continual learning, scientific literature synthesis, hallucination, citation verification, and research-agent evaluation.

2. Background and Related Work

2.1 Parametric knowledge and the knowledge cutoff

Modern LLMs store statistical representations of language and world knowledge in their parameters. This design gives them broad generalization capabilities but creates a fundamental separation between model knowledge and current external knowledge. Lewis et al. (2020) characterize this distinction through parametric and non-parametric memory: the former is encoded in model parameters, while the latter can be accessed dynamically through retrieval.

A nominal knowledge cutoff is therefore best understood as an approximation to the latest point in time represented by training data. Cheng et al. (2024) introduce the concept of an effective cutoff, showing that the temporal boundary of knowledge can vary by source and topic. This is especially relevant to literature reviews because different scientific subfields have different publication patterns and different representation in web-scale corpora.

A model may consequently possess detailed knowledge of an older research area while having little or

no representation of a newly emerging subfield. Conversely, some newer facts may appear in its training distribution even when its nominal cutoff is earlier. This makes the statement "the model's knowledge ends on date X" an oversimplification.

2.2 Temporal knowledge and evolving facts

The temporal dimension of LLM knowledge has become an active research area. Kim et al. (2024), through the EvolvingQA benchmark, demonstrate that models struggle to update and remove outdated knowledge and are particularly challenged by numerical and temporal changes.

Zhang and Choi (2023) frame this problem as temporal misalignment: a model trained on historical data may be asked questions about the present. They propose predicting the duration for which facts remain valid and show that identifying volatile facts can improve calibration and reduce the use of outdated information.

More recent work has expanded evaluation beyond static factual recall. EvolveBench evaluates temporal competence through cognition, temporal awareness, trustworthiness, understanding, and reasoning. Its evaluation of 15 widely used LLMs found that even strong models continue to struggle with temporally misaligned contexts (Zhu et al., 2025).

These findings suggest that temporal reliability cannot be reduced to model size or general knowledge. A model may be highly capable at reasoning while still failing to determine whether a particular claim is valid at the requested point in time.

2.3 Hallucination and citation reliability

Knowledge cutoffs interact strongly with hallucination. A model that cannot retrieve a recent paper may generate a plausible but unsupported description of what the literature supposedly contains. More generally, LLM-generated text can contain both intrinsic and extrinsic hallucinations, including unsupported or factually inconsistent information (Ji et al., 2023).

The problem is particularly serious for academic writing because literature reviews require citations. Walters and Wilder (2023) evaluated ChatGPT-generated literature reviews across 42 multidisciplinary topics and identified fabricated and erroneous bibliographic references.

This produces an important distinction between citation existence and citation validity. A citation may correspond to a real paper but still fail to support the statement attached to it. Conversely, an LLM may cite an older, genuine paper because it cannot access a newer study, producing a formally valid but temporally incomplete review.

Therefore, the knowledge-cutoff problem has at least three dimensions:

- Coverage error: important recent literature is missing.
- Temporal attribution error: a valid source is presented as representing current knowledge when it represents an earlier state.
- Citation fabrication/error: the model produces nonexistent or inaccurate references.

2.4 LLMs for literature-review workflows

Research on LLM-assisted literature review increasingly moves beyond simple summarization. A 2025 scoping review of 37 studies found that LLMs had been applied to 10 of 13 stages of systematic-review

workflows, with literature search, study selection, and data extraction among the most frequent applications. However, fully validated applications remained uncommon (Wagner et al., 2025).

Tools such as LitLLM use web search, LLM-based keyword generation, reranking, and RAG to construct related-work sections. The system was explicitly motivated by the limitations of conventional LLMs, including hallucination and failure to incorporate research released after model training (Agarwal et al., 2024).

PaperQA similarly demonstrates how retrieval across full-text scientific papers can support evidence-grounded question answering and synthesis. Its LitQA benchmark was designed to test retrieval and synthesis across multiple scientific papers, illustrating a transition from closed-book generation toward evidence-grounded research assistance (Lála et al., 2023).

3. Main Technical Discussion: How Knowledge Cutoffs Distort Literature Reviews

3.1 Incomplete temporal coverage

The most direct consequence of a knowledge cutoff is missing literature. Suppose a rapidly evolving field publishes 30% of its relevant literature during the year following a model's training cutoff. A closed-book LLM cannot reliably incorporate those publications unless they are supplied in context or accessed through external retrieval.

The resulting review may nevertheless appear comprehensive because language fluency masks the absence of evidence. This is more dangerous than an obviously incomplete answer because researchers may mistake stylistic coherence for scholarly coverage.

In rapidly evolving fields, recent papers are disproportionately important because they may introduce:

- new benchmark results;
- newly discovered limitations;
- revised methodologies;
- changes in regulatory or ethical consensus;
- new datasets;
- competing approaches;
- replications or negative findings; and
- corrections to previously accepted conclusions.

Consequently, omission is not temporally neutral. Missing a recent paper can change the direction of the review.

3.2 Citation-lag bias

Scientific influence is not synchronized with publication date. New papers often have few citations, while older papers accumulate many citations over time. Automated literature systems that use citation counts or citation networks as proxies for importance can therefore systematically favor older literature.

Wu et al. (2025) explicitly identify citation lag as a problem for automated review generation in rapidly evolving fields. Their system was designed partly to avoid underestimating recent publications.

This produces a citation-lag bias:

$$P(\text{paper retrieved}) \neq P(\text{paper relevance})$$

Instead, retrieval probability may depend on age, citation count, indexing history, visibility, and textual prominence. A literature review can consequently become "popular-paper-centric" rather than "state-of-the-art-centric."

3.3 Stale consensus

A more subtle effect occurs when the LLM has learned an older consensus. Suppose a field historically concluded that Method A was superior to Method B. Several newer papers subsequently demonstrate that Method B performs better under realistic conditions. A cutoff-limited LLM may continue to describe Method A as state of the art.

This is not necessarily a hallucination. The model may generate a historically accurate statement using genuine sources. The error lies in its temporal framing.

The distinction can be formalized as:

$$\text{Factual correctness} \neq \text{Temporal correctness}.$$

A claim can be true at time t_1 and false or incomplete at t_2. Literature-review systems therefore require evaluation against both dimensions.

3.4 Distorted research gaps

Identifying research gaps is one of the most valuable functions of a literature review, but it is also one of the functions most vulnerable to cutoff effects.

If an LLM lacks recent publications, it may state:

> "Few studies have investigated X."

when in reality several studies published after its cutoff have investigated X extensively.

This creates false novelty. Researchers may pursue questions that have already been addressed, duplicate experiments, or frame their contributions as novel when they are not.

The reverse problem is also possible. A model may identify a historical gap that recent work has already addressed, thereby understating the progress of a field.

3.5 Misinterpretation of trends

A literature review is fundamentally temporal. It should answer questions such as:

- What methods dominated in earlier research?
- What changed recently?
- Why did the change occur?

- Which findings have been replicated?
- Which approaches are emerging?
- What remains uncertain?

A static LLM is poorly suited to these questions unless it has explicit temporal evidence.

This is why the emerging literature on temporal awareness is important. EvolveBench shows that even capable models struggle with temporal misalignment, while Kim et al.'s EvolvingQA demonstrates weaknesses in updating and removing obsolete knowledge (Kim et al., 2024; Zhu et al., 2025).

3.6 The problem of changing terminology

Rapidly evolving fields frequently change terminology. New concepts may replace older terms, while a single concept may be referred to differently by different research communities.

A cutoff-limited model may therefore:

1. fail to recognize a new term;
2. interpret a new term using an older definition;
3. miss papers because retrieval queries use obsolete terminology; or
4. incorrectly merge historically distinct concepts.

This is particularly consequential in interdisciplinary fields, where terminology evolves faster than indexing systems.

3.7 Conflicting temporal evidence

Adding retrieval does not automatically solve the problem. A retrieval corpus may contain multiple versions of the same information: preprint, conference paper, journal article, correction, retraction, updated guideline, or later replication.

A model must determine not merely which source is relevant, but which source is temporally authoritative.

The HoH benchmark demonstrates this issue directly. Ouyang et al. (2025) found that outdated information in a retrieval knowledge base can reduce accuracy and mislead models even when current information is also available.

Thus, RAG transforms the problem from:

> "Does the model know the latest information?"

into:

> "Can the system retrieve, identify, rank, reconcile, and correctly attribute information according to time?"

4. Current Approaches and Developments

4.1 Retrieval-augmented generation

RAG is the most established solution to knowledge staleness. Instead of relying entirely on parametric memory, the system retrieves external documents and conditions generation on them (Lewis et al., 2020).

For literature reviews, a basic RAG pipeline can be represented as:

$$Q \rightarrow R(Q, D) \rightarrow C \rightarrow G(C, Q)$$

where Q is the research question, D is the external corpus, R is retrieval, C is retrieved context, and G is generation.

RAG allows the underlying LLM to remain unchanged while the evidence corpus is updated. However, retrieval quality becomes a critical dependency.

4.2 Scholarly RAG

General web search is not necessarily sufficient for scholarly review generation. Scientific systems need metadata, abstracts, full text, publication dates, versions, citations, author information, and preferably persistent identifiers.

PaperQA represents an early research-oriented RAG approach, retrieving full-text scientific literature and synthesizing evidence with citations (Lála et al., 2023).

OpenScholar represents a more recent development. Asai et al. (2026) introduced a specialized retrieval-augmented language model backed by a datastore containing approximately 45 million open-access papers. The system combines specialized retrieval, reranking, and iterative self-feedback. Its evaluations found substantial improvements over conventional LLM-only approaches and strong performance in citation accuracy and scientific-answer correctness.

The significance of OpenScholar is not simply higher retrieval accuracy. It demonstrates that architecture and evidence infrastructure can be as important as model scale for scientific synthesis.

4.3 Temporal filtering

A simple but powerful strategy is to make time explicit during retrieval. A query can specify:

$$D_t = \{d \in D \mid publication_date(d) \leq t\}$$

For a "state of the field as of 2023" review, documents published after 2023 should not be retrieved. Conversely, a current review should prioritize documents close to the present.

Temporal filtering can be combined with relevance:

$$Score(d) = \alpha R(d, q) + \beta T(d, t) + \gamma A(d) + \delta C(d)$$

where:

- R = semantic relevance;

- T = temporal relevance;
- A = source authority;
- C = citation or scholarly significance.

Such scoring prevents a highly cited but obsolete paper from automatically outranking a recent and directly relevant study.

4.4 Continual learning

Another approach is to update model parameters periodically. Continual-learning research distinguishes continual pretraining, domain-adaptive pretraining, and continual fine-tuning (Wu et al., 2024; Shi et al., 2024).

The advantage is that frequently needed new knowledge becomes part of the model's internal representation. The disadvantage is cost and the risk of catastrophic forgetting.

For literature-review systems, continual training also introduces a reproducibility problem: if a model changes continuously, two researchers querying the "same" system at different times may obtain different reviews.

Thus, continuously updated models may improve freshness while reducing experimental reproducibility.

4.5 Knowledge editing

Knowledge editing modifies specific model representations rather than retraining the entire system. Temporal knowledge editing is especially relevant when facts become outdated.

Yin et al. (2024) argue that outdated knowledge should not necessarily be erased. Historical information may remain valuable when a user asks a historical question. Their work distinguishes correcting intrinsic model errors from integrating new knowledge that supersedes previous knowledge.

This distinction is essential for literature reviews. A good review must preserve the history of a field rather than simply overwrite old knowledge.

4.6 Fact-duration modeling

Zhang and Choi (2023) propose predicting how long facts remain valid. Their results indicate that volatile facts should trigger stronger reliance on external sources.

A literature-review system could extend this principle from facts to claims. For example:

- established mathematical theorem → low temporal volatility;
- drug efficacy estimate → medium/high volatility;
- state-of-the-art benchmark result → high volatility;
- current regulatory guidance → very high volatility.

This suggests a promising design principle: retrieval effort should be proportional to expected temporal volatility.

4.7 Agentic deep research

Recent systems increasingly use agents that plan searches, execute multiple retrieval steps, compare sources, and generate structured reports. DeepResearch Bench evaluates such systems using 100 PhD-level tasks spanning 22 fields and emphasizes both research quality and citation accuracy.

Agentic architectures can reduce cutoff effects because they can access information published after the model's training cutoff. However, they also introduce additional failure modes:

- query drift;
- source-selection bias;
- retrieval loops;
- incomplete search coverage;
- premature stopping;
- inconsistent source ranking;
- opaque decision processes; and
- temporal contamination.

Consequently, "agentic" does not automatically mean "systematic."

4.8 Citation verification

Citation verification is an increasingly important safeguard. Walters and Wilder (2023) demonstrate that LLM-generated literature reviews can contain fabricated references, while recent research continues to document reference hallucination in academic-style outputs.

Verification should occur at multiple levels:

1. Existence: Does the paper actually exist?
2. Metadata: Are authors, title, year, venue, and identifier correct?
3. Temporal validity: Was the paper available by the relevant cutoff?
4. Entailment: Does the cited paper support the claim?
5. Scope: Does the cited evidence support the strength of the wording?
6. Completeness: Are major competing or contradictory studies missing?

OpenScholar's evaluation illustrates the value of treating citation accuracy as a first-class metric rather than an afterthought.

5. Research Challenges and Limitations

5.1 No universal definition of "current"

A literature review can be current relative to:

- the publication date of the review;
- the last database search;

- a specified historical cutoff;
- the latest peer-reviewed literature;
- the latest preprints;
- a field-specific guideline;
- or a dynamically changing evidence base.

Therefore, temporal correctness requires an explicit reference time.

5.2 Preprints and publication versions

Rapidly evolving fields often rely heavily on preprints. A paper may exist as a preprint in January, a conference paper in June, and a journal version the following year.

A temporally accurate system must represent these as related versions rather than independent evidence.

5.3 Retractions and corrections

Scientific knowledge is not monotonically additive. New papers can invalidate earlier results through corrections, retractions, failed replications, or improved methodologies.

A simple "latest paper wins" strategy is therefore inadequate. Temporal reasoning must be combined with evidence quality and scientific authority.

5.4 Retrieval bias

RAG systems inherit biases from their retrieval corpus. If the corpus underrepresents certain journals, languages, geographic regions, disciplines, or publication types, the generated review will inherit those gaps.

This means that retrieval can improve freshness without improving representativeness.

5.5 Conflicting evidence

Scientific fields rarely progress through simple replacement. Competing studies may coexist for years.

An LLM-generated review may collapse disagreement into a single narrative because language generation rewards coherence. This creates a tension between readability and epistemic fidelity.

5.6 Citation correctness versus citation completeness

A review can have 100% valid citations and still be poor if it omits major recent studies.

This suggests that evaluation should distinguish:

$$\text{Citation Accuracy}$$

from

$$\text{Evidence Coverage}.$$

Both are necessary for scholarly reliability.

5.7 Evaluation benchmarks remain incomplete

Recent benchmarks such as EvolveBench and HoH are significant advances, but most evaluations still focus on question answering rather than full literature-review generation.

A literature review requires broader evaluation dimensions:

- search completeness;
- chronological coverage;
- citation accuracy;
- claim-evidence alignment;
- synthesis quality;
- disagreement representation;
- methodological comparison;
- gap identification;
- and reproducibility.

5.8 Reproducibility

Dynamic retrieval creates a paradox. A system can be more current but less reproducible.

If the same query is executed six months apart, the retrieved literature will differ. Therefore, scholarly use requires preservation of:

- query strings;
- search date;
- database version;
- retrieved document IDs;
- document versions;
- ranking results;
- model version;
- prompts;
- and generation settings.

Without this information, an AI-generated review cannot easily be audited.

6. Research Gaps and Future Directions

6.1 Temporal completeness as a formal evaluation metric

Current evaluation commonly emphasizes factual correctness and citation accuracy. Future literature-review benchmarks should measure temporal completeness.

One possible metric is:

$$TC = \frac{\sum_{i=1}^{n} w_i I_i}{\sum_{i=1}^{n} w_i}$$

where $I_i = 1$ if a substantively important publication or finding is represented and w_i reflects its relevance or importance.

This would distinguish a review that is factually correct but incomplete from one that provides broad current coverage.

6.2 Claim-level temporal provenance

Every substantive claim should ideally have a provenance tuple:

$$P = (\mathit{source}, \mathit{date}, \mathit{version}, \mathit{evidence}, \mathit{confidence})$$

This would enable a system to state not simply "Method X is effective," but:

> "Studies published between 2024 and 2026 generally report improved performance under conditions A and B, although evidence remains mixed under condition C."

Such statements better represent scientific uncertainty and temporal evolution.

6.3 Temporal knowledge graphs

Future systems could represent literature as a temporal graph:

$$G = (V, E, T)$$

where nodes represent papers, methods, datasets, claims, and findings; edges represent citations, comparisons, replications, contradictions, and extensions; and T records temporal validity.

This would allow the LLM to reason over changes in relationships rather than merely retrieve documents.

6.4 Temporal conflict resolution

Systems need explicit mechanisms for deciding what to do when sources disagree.

A useful hierarchy might consider:

1. publication date;
2. peer-review status;
3. methodological quality;
4. replication evidence;
5. retraction/correction status;
6. sample size;
7. domain authority; and
8. directness of evidence.

The latest source should not automatically dominate.

6.5 Time-aware research agents

Future deep-research agents should maintain a temporal state during search. Rather than simply accumulating sources, the agent should ask:

- When was this claim established?
- Has it subsequently changed?
- Is there newer contradictory evidence?
- Was this paper available by the requested cutoff?
- Is this a preprint or final publication?
- Has this result been replicated?

This would transform research agents from document collectors into temporal evidence synthesizers.

6.6 Benchmarking literature reviews rather than individual answers

The field needs benchmark suites specifically designed for AI-generated literature reviews. Such benchmarks should contain rapidly changing topics with known publication timelines and expert-created reference reviews.

Evaluation should measure:

Dimension	Core question
Temporal coverage	Did the system find important recent work?
Citation validity	Do cited papers exist and have correct metadata?
Citation entailment	Does each citation support the claim?
Temporal attribution	Is evidence placed in the correct historical context?
Trend fidelity	Does the review accurately describe field evolution?
Conflict representation	Are competing findings represented?
Gap validity	Are claimed gaps actually unresolved?
Reproducibility	Can another researcher reconstruct the review?

6.7 Human-AI collaboration

The evidence increasingly supports treating LLMs as research assistants rather than autonomous replacements for scholarly judgment. A 2025 scoping review concluded that LLM applications in systematic reviews are promising but not yet sufficiently validated for unrestricted use.

Human researchers remain particularly important for:

- assessing methodological quality;
- interpreting contradictory findings;
- judging novelty;
- identifying disciplinary context;
- evaluating source credibility;
- and determining whether a research gap is genuinely meaningful.

The optimal workflow is therefore likely to be human-directed, machine-accelerated synthesis.

6.8 Explicit uncertainty and abstention

An LLM should be able to say:

> "The available evidence after 2025 is insufficient to establish a consensus."

This is preferable to generating a confident synthesis from incomplete evidence.

Recent research on prompting earlier knowledge cutoffs shows that models can be instructed to simulate temporal restrictions but struggle to prevent leakage from causally related post-cutoff information (Gao et al., 2025).

This demonstrates why temporal awareness cannot depend exclusively on prompting. The system must have mechanisms for verifying temporal admissibility.

7. Conclusion

Knowledge cutoffs create a fundamental reliability problem for LLM-generated literature reviews in rapidly evolving scientific fields. The problem is not limited to missing recent papers. A static knowledge boundary can influence the perceived state of the field, distort citation patterns, preserve obsolete consensus, generate false research gaps, and misrepresent emerging methodological trends.

The central insight is that scholarly synthesis is inherently temporal. A literature review describes not only what is known, but when it became known, how evidence changed, which findings were superseded, and where uncertainty remains. Conventional LLMs are optimized primarily for linguistic prediction and general reasoning, not for maintaining an auditable temporal model of scientific knowledge.

RAG provides an important architectural response by separating language-generation capabilities from external knowledge storage. Scholarly systems such as PaperQA, LitLLM, automated review-generation pipelines, and OpenScholar demonstrate increasingly sophisticated ways of combining retrieval with LLM-based synthesis. Nevertheless, retrieval introduces its own temporal problems: outdated documents can coexist with current ones, citation networks can privilege older literature, and retrieval systems can fail to distinguish different versions or conflicting findings.

The emerging research on temporal benchmarks reinforces this conclusion. EvolvingQA, EvolveBench, and HoH demonstrate that temporal awareness and outdated-information handling remain unresolved even for highly capable models.

Future literature-review systems should therefore treat time as a first-class property of evidence. Reliable systems should combine timestamp-aware scholarly retrieval, temporal provenance, claim-level citation verification, source-version tracking, conflict resolution, uncertainty estimation, and reproducible search logs. Evaluation should expand beyond factuality toward temporal completeness, trend fidelity, gap validity, and historical attribution.

The ultimate goal should not be an LLM that merely produces a review that sounds current. It should be a research system that can demonstrate why its review is current, which evidence supports each claim, when that evidence was available, what evidence contradicts it, and what remains unknown. Such capabilities would transform LLM-generated literature reviews from fluent summaries into auditable, temporally grounded

instruments for scientific knowledge synthesis.

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